

RAJMUDRA FINTECH

rajmudrafintech.com

SYSTEM REQUIREMENTS SPECIFICATION

Trading Education Platform — Full Stack MERN Application

Document Version	1.0
Date	April 2026
Platform	rajmudrafintech.com
Stack	MERN (MongoDB, Express, React, Node.js)
Payment Gateway	Cashfree
Live Classes	Google Meet
Streaming	YouTube Unlisted (Behind Login)
Status	Draft — Pending Review

PREPARED BY: DEVELOPMENT TEAM | CONFIDENTIAL

1. Introduction

1.1 Purpose

This System Requirements Specification (SRS) defines the complete functional, non-functional, and technical requirements for the Rajmudra Fintech web application — a full-stack MERN trading education platform hosted at rajmudrafintech.com. The platform enables student enrolment, Cashfree-powered payments, Google Meet live classes, YouTube-embedded lecture delivery, user dashboards, and a multi-tier admin system.

1.2 Scope

The Rajmudra Fintech platform serves as a premium Forex trading education system founded and mentored by Vishal Tompe Patil and Nilesh Thore Patil. It replaces the existing static HTML website with a dynamic, authenticated, payment-enabled full-stack application while preserving the existing brand identity: dark luxury theme (black #080808, gold #C9A84C), Playfair Display + Montserrat typography, and all current marketing content.

1.3 Technology Stack

Layer	Technology	Purpose
Frontend	React.js + Vite	SPA with component-based UI
Backend	Node.js + Express.js	REST API server
Database	MongoDB + Mongoose	Document-based data store
Authentication	JWT + bcrypt	Secure session management
Payments	Cashfree Payment Gateway	INR payments — UPI, Card, Net Banking, EMI
Live Classes	Google Meet (link delivery)	Live trading sessions via secure dashboard link
Video Content	YouTube Unlisted Embed	Lecture delivery behind authentication wall
State Management	Redux Toolkit / Zustand	Global frontend state
Styling	Tailwind CSS + Custom CSS	Responsive design matching existing brand
Email	Nodemailer / SendGrid	Transactional emails
Hosting	AWS / DigitalOcean / Render	Production deployment

Layer	Technology	Purpose
Analytics	Google Analytics 4	Traffic and conversion tracking

1.4 User Roles & Hierarchy

The system operates with four distinct user roles, each with clearly defined access levels:

Role	Description	Access Level
Super Admin	Platform owner (you) — full system access	Maximum — can manage everything
Admin (Mentor 1)	Vishal Tompe Patil — primary mentor and founder	High — manage students, content, live sessions
Admin (Mentor 2)	Nilesh Thore Patil — second mentor and founder	High — manage students, content, live sessions
Student/User	Enrolled paying student	Limited — own dashboard, purchased courses only

1.5 Document Conventions

- FR-xx = Functional Requirement
- NFR-xx = Non-Functional Requirement
- UC-xx = Use Case
- UI-xx = UI/UX Specification
- All monetary values are in Indian Rupees (INR) inclusive of GST
- Priority levels: Critical > High > Medium > Low

2. Existing Static Website Analysis

2.1 Design Identity (Must Be Preserved)

The MERN application must exactly replicate the visual identity of the current static site. All design tokens below are mandatory.

Design Token	Value	Usage
Primary Background	#080808 (--black)	Main page background
Deep Background	#0e0e0e (--deep)	Section backgrounds
Card Background	#131313 (--card)	Card and panel backgrounds
Border Color	#232323 (--border)	Dividers and borders

Design Token	Value	Usage
Gold Primary	#c9a84c (--gold)	Primary accent — headings, CTAs, icons
Gold Light	#e8c96a (--gold-light)	Hover states on gold elements
Gold Pale	#f5e5b0 (--gold-pale)	Subtle gold highlights
Silver	#b8b8c8 (--silver)	Secondary text, subheadings
White	#f0ede8 (--white)	Primary text on dark background
Muted	#888880 (--muted)	Tertiary text, labels, metadata
Accent Red	#c0392b (--accent-red)	Error states, warnings
Primary Font	Playfair Display (serif)	Headlines, hero text, large display
Secondary Font	Cormorant Garamond (serif, italic)	Subheadings, pull quotes
Body Font	Montserrat (sans-serif, weight 300–700)	Body text, labels, buttons
Button Shape	clip-path polygon (cut corners)	Primary CTA buttons

2.2 Existing Page Sections (All Must Be Retained)

Section	Content Summary	Notes
Market Ticker	Scrolling ticker: DXY, USOIL, XAUUSD, XAGUSD, BTC, ETH, Forex pairs, US stocks	Animated, mock data for now; future API integration planned
Hero Section	"Trade with the Precision of Institutions" — CTAs to Programme & Contact	Full-viewport, radial gold gradient background with grid overlay
Stats Bar	5 Phases, 4 Days Live, 100% Dedication, Infinite Audit	4-column grid with Playfair Display numerals
Highlights Grid	6 cards: Dedicated, Roadmap, 4-Day Live, Power of Three, Handholding, Auditing	Gold left-border reveal on hover
Syllabus Section	5 phases: Foundation, Technical Analysis, Advanced Strategy, Risk Mgmt, Execution	Dark background, 5-column phase cards with tag chips
Who Is This For	Admission criteria, tagline quote, 500+ students badge	Gold-bordered card with left gold accent line
Mentors Section	Vishal Tompe Patil (VTP) + Nilesh Thore Patil (NTP) with bios and WhatsApp CTAs	Two mentor cards with initials badge and WhatsApp buttons
Testimonials	3 testimonials: Madhav (Lawyer), Abhishek (Trader), Prashil (Investor)	Dark cards, first-letter avatar circles

Section	Content Summary	Notes
Contact Section	Phone (+91 70204 08147, +91 93704 73645), Email, WhatsApp keyword TRADER	Two-column layout — details left, form/CTA right
Footer	Logo, tagline, quick links, contact info, legal links	4-column grid, Terms & Privacy modals
Terms Modal	Full T&C including FEMA compliance, refund policy, code of conduct	Overlay modal with scrollable legal content
Privacy Modal	Privacy policy including GA4 disclosure, data handling, cookie policy	Overlay modal with scrollable legal content

3. Functional Requirements

3.1 Public-Facing Website (Unauthenticated)

All sections from the existing static site must be preserved with identical design, content, and animations.

FR-01: Complete Static Content Preservation

All sections of the existing website must be rebuilt as React components with identical visual design, typography, colors, animations, and content. This includes the market ticker, hero, stats bar, highlights grid, syllabus, mentors, testimonials, contact section, footer, and legal modals.

FR-02: Animated Market Ticker

Property	Specification
Symbols	DXY, USOIL, XAUUSD, XAGUSD, BTC, ETH, USD/JPY, EUR/USD, GBP/USD, AUD/USD, TSLA, GOOGL, MSFT, META, NVDA, AAPL
Animation	Continuous left-scroll, 35s loop, pauses on hover (animation-play-state: paused)
Color Coding	Green (#2ecc71 bg rgba) for positive change, Red (#e74c3c bg rgba) for negative
Font	Times New Roman for price/change values, Montserrat for symbols
Data	Mock static data on launch; Forex/stock API integration in Phase 2

FR-03: Hero Section CTA Flow

- 'Explore the Programme' button scrolls smoothly to #syllabus section
- 'Enrol Now' / 'Get Started' button (to be added) opens signup/registration flow
- 'Contact Us' button scrolls to #contact section
- WhatsApp chat links open wa.me with pre-filled messages as per existing site

FR-04: Enrolment CTA Integration

A prominent 'Enrol Now' button must be added to the hero section and course pricing section, which redirects authenticated users to payment checkout and unauthenticated users to the registration page.

FR-05: Course Pricing Page

Field	Value
Programme Name	Professional Forex Mastery
Price	INR 39,999 (inclusive of GST)
Badge	"Most Popular" / "Limited Seats"
Features Listed	5 Phases, 4-Day Live Trading, Trade Audit, Mentor Access, Lifetime Access
Languages	English + Hindi
CTA	Enrol Now — triggers Cashfree payment

3.2 Authentication System

FR-06: Student Registration

Field	Type	Validation
Full Name	Text	Required, 2-100 characters
WhatsApp Number	Tel	Required, Indian 10-digit, validated format
Email Address	Email	Required, valid email format, unique in DB
Password	Password	Required, min 8 chars, 1 uppercase, 1 number
Confirm Password	Password	Must match password field
Interest	Select	Required — Course / Signals / Mentorship / VIP
Terms Agreement	Checkbox	Required — must accept T&C and Privacy Policy

- On successful registration: account created with role='student', status='pending_payment'
- Verification email sent with email confirmation link
- User redirected to pricing/checkout page
- WhatsApp confirmation message sent (optional, Phase 2)

FR-07: Student Login

- Login via Email + Password
- JWT token issued on success (15-minute access token + 7-day refresh token)
- Tokens stored in httpOnly cookies (not localStorage for security)
- 'Remember Me' option extends refresh token to 30 days
- 'Forgot Password' sends reset link to registered email (expires in 1 hour)
- Rate limiting: 5 failed attempts locks account for 15 minutes

FR-08: Admin Login

- Separate /admin/login route with dedicated admin login page
- Two-Factor Authentication (2FA) via OTP sent to registered mobile (Phase 2 — initially email OTP)
- Session timeout: 8 hours of inactivity
- All admin actions logged with timestamp, IP, and user ID

FR-09: Super Admin Login

- Separate /superadmin/login route
- Mandatory 2FA via email OTP (enforced, not optional)
- IP whitelisting option (configurable in settings)
- All actions logged with full audit trail

3.3 Payment Integration — Cashfree

FR-10: Cashfree Checkout

Integration with Cashfree Payment Gateway for processing INR 39,999 course enrolments.

Feature	Specification
Gateway	Cashfree Payments — Production API
Supported Methods	UPI (GPay, PhonePe, Paytm), Debit/Credit Card, Net Banking, EMI, Wallets
Currency	INR only
Amount	INR 39,999 inclusive of GST
Order ID Format	RMF-{YYYY}-{random 8-char alphanumeric}
Payment Status Webhook	Cashfree webhook endpoint: POST /api/webhooks/cashfree
Success Flow	Webhook confirms payment → student status set to 'active' → welcome email sent → dashboard unlocked
Failure Flow	Payment failed notification → retry option → support contact

Feature	Specification
Refund Policy	No refunds (per T&C) — admin can manually process exceptions via Super Admin panel

FR-11: Invoice Generation

- Auto-generate GST-compliant PDF invoice on successful payment
- Invoice includes: Order ID, student name, email, phone, course name, amount, GST breakdown, date
- Invoice emailed to student and stored in database
- Downloadable from student dashboard

3.4 Student Dashboard (Authenticated — Post-Payment)

The student dashboard must match the existing brand identity: dark black/gold luxury theme. Layout should feel like an exclusive members-only portal.

FR-12: Dashboard Home / Overview

UI Element	Description
Welcome Banner	Personalised greeting: 'Welcome back, [Name]' with gold accent
Progress Bar	Course completion percentage with Playfair Display numerals
Quick Stats	4-card strip: Phase Progress, Next Live Session countdown, Messages, Support Tickets
Upcoming Session	Next Google Meet session date/time with 'Join Now' button (revealed 15 min before)
Announcements	Latest 3 announcements from admin — badge indicates unread count
Quick Links	Phases, Live Classes, Materials, Support — icon grid

FR-13: Course Phases & Content

- 5 collapsible phase cards matching syllabus design (Foundation, Technical, Advanced, Risk, Execution)
- Each phase unlocks sequentially (admin can override)
- Phase status: Locked / In Progress / Completed — shown with gold/grey indicators
- Materials per phase: PDF downloads, notes (uploaded by admin)
- YouTube lecture videos embedded per phase (iframe embed, unlisted URL stored in DB)
- Mark complete checkbox per phase — tracks progress

FR-14: Live Classes (Google Meet)

Feature	Description
Schedule Display	Calendar view showing upcoming live sessions with date, time, topic, mentor name
Join Button	Active 15 minutes before session — opens Google Meet link in new tab
Link Security	Google Meet link stored in DB, revealed only to active students on schedule time
Session History	Past sessions listed with date, topic, mentor — no replay (recordings not in Phase 1)
Timezone	IST (Indian Standard Time) — displayed with IST label
Mentor Indicator	Shows whether session is with Vishal or Nilesh

FR-15: My Profile

- View and edit: Name, Phone, Email, Profile Photo (avatar upload)
- Change password with current password verification
- View enrolment date, payment receipt, invoice download
- WhatsApp number update — requires admin approval
- Language preference: English / Hindi

FR-16: Support / Query System

- Student can raise a support ticket with Subject, Category (Technical/Course/Payment/Other), and Message
- Ticket assigned auto-ticket number: RMF-TKT-XXXXXX
- Student sees ticket status: Open / In Progress / Resolved
- Reply thread visible in dashboard (admin replies appear here)
- Email notification on every status change

FR-17: Announcements

- Admins can post announcements (text, optionally with a PDF attachment)
- Students see latest announcements in dashboard — unread badge on nav
- Announcements can be tagged: General / Course Update / Live Session / Important

FR-18: Trade Audit Submission

- Student can submit a trade screenshot/PDF for review
- Form: Date, Pair/Instrument, Direction (Buy/Sell), Entry Price, Exit Price, Result (P/L), Notes
- Admin reviews and posts written feedback
- Student receives email notification when feedback is posted

3.5 Admin Dashboard — Vishal Patil & Nilesh Thore Patil

Both Vishal and Nilesh have Admin role. Both can perform the same admin functions unless restricted by Super Admin.

FR-19: Admin Home / Overview

Widget	Data Shown
Total Students	Count of all active enrolled students
Revenue MTD	Month-to-date Cashfree payments total (INR)
Pending Queries	Open support tickets count — red badge if > 5
Upcoming Sessions	Next scheduled Google Meet session
New Registrations	Students registered in last 7 days
Trade Audits Pending	Unreviewed trade audit submissions

FR-20: Student Management

- Table view: Name, Email, Phone, Enrolment Date, Payment Status, Phase Progress, Last Active
- Search by name / email / phone
- Filter by: Status (Active/Pending/Suspended), Phase, Batch, Enrolment Date range
- View individual student profile with full details and activity log
- Manually activate / suspend / reactivate a student account
- Send direct WhatsApp message to individual student (pre-filled URL)
- Export student list as CSV/Excel

FR-21: Course Content Management

- Upload/edit/delete course materials (PDF, notes) per phase
- Add YouTube unlisted URL for each lecture video per phase
- Lock / unlock individual phases per student or globally
- Bulk unlock phases for all students (e.g., advance whole batch)

FR-22: Live Session Management

- Schedule a live session: Date, Time (IST), Topic, Mentor (Vishal/Nilesh), Google Meet Link, Description
- Edit or cancel a scheduled session — students notified via email on cancellation
- Session goes 'live' automatically (Join button activates) 15 minutes before scheduled time

- Mark session as completed after it ends
- Session history with attendance tracking (manual checkbox per student, Phase 2: auto)

FR-23: Announcement Management

- Create new announcement: Title, Body, Tag (General/Course/Live/Important), Attachment (optional PDF)
- Edit / delete announcements
- Send email blast to all active students for 'Important' tagged announcements

FR-24: Support Ticket Management

- View all open tickets sorted by priority (Important first) and date
- Filter by: Status, Category, Student
- Reply to a ticket — student notified via email
- Change ticket status: Open → In Progress → Resolved
- Close a ticket with resolution note

FR-25: Trade Audit Review

- List of pending trade audit submissions with student name, date, instrument
- View uploaded screenshots and student notes
- Post written feedback per trade audit
- Mark audit as reviewed — student notified via email

3.6 Super Admin Dashboard (Platform Owner)

Super Admin is you — the platform owner. This dashboard provides complete visibility and control over the entire system.

FR-26: Super Admin Home

Metric	Description
Total Revenue	All-time and MTD Cashfree payments — with trend chart
Total Students	Active / Pending / Suspended count
System Health	API status, DB connection, Cashfree webhook status
Recent Admin Activity	Last 20 actions performed by Vishal and Nilesh with timestamps
Failed Payments	Count and list of failed/abandoned payment attempts
Support Overview	Open tickets by category — donut chart

FR-27: Full User Management

- View, search, filter ALL users including admins
- Create new Admin accounts (Mentor 1, Mentor 2, or future admins)
- Reset any user's password
- Permanently delete a user account with data cascade options
- Promote/demote user roles (Student ↔ Admin) — Super Admin cannot be demoted by anyone
- View complete activity log per user

FR-28: Payment & Revenue Management

- Full payment history: all transactions with Order ID, Student, Amount, Status, Date, Method
- Filter by: Date range, Status (Success/Failed/Refunded), Payment Method
- Manual payment entry (for bank transfer enrolments)
- Export payment reports as CSV/Excel/PDF
- Cashfree webhook log — last 100 webhook events with payload
- Initiate refund (manual — generates refund record, actual refund via Cashfree dashboard)

FR-29: Query Resolution Center

- View ALL support tickets from ALL students across both admin accounts
- Reassign a ticket from one admin to another
- Escalate a ticket and flag for priority resolution
- View full ticket thread and add internal notes (not visible to student)
- Generate support report: tickets by category, average resolution time

FR-30: Platform Settings

- Update site-wide settings: Platform name, WhatsApp numbers, Email configuration
- Manage announcement templates
- Configure course pricing (requires confirmation step to change INR 39,999)
- Toggle maintenance mode — displays maintenance page to students, admins still have access
- Manage email templates (welcome, payment success, password reset, ticket update)
- View and manage error/system logs

FR-31: Admin Activity Audit Log

Every action performed by any admin (Vishal or Nilesh) is logged with:

- Timestamp (IST)
- Admin name and role
- Action performed (e.g., 'Unlocked Phase 3 for student Raj Sharma')
- IP address and device/browser info

- Affected entity (student ID, ticket ID, etc.)
- Super Admin can filter, search, and export logs

4. UI/UX Specifications

4.1 Design System

All UI components must follow the existing brand design system extracted from the static site. The following specifications are mandatory for all new screens.

UI-01: Navigation (Public)

Element	Specification
Logo	Rajmudra Fintech logo — gold glow drop-shadow filter (existing logo file)
Nav Links	Home, Features, Courses, Community — Montserrat 600, 11px, letter-spacing 2px, uppercase
WhatsApp Icon	Font Awesome, links to wa.me/917020408147 with pre-filled message
Login Button	Outline style — gold border, gold text, polygon clip-path
Enrol Now Button	Primary style — gold gradient fill, black text, polygon clip-path, hover: lift + glow
Sticky Behavior	Sticky on scroll — background darkens to <code>rgba(8,8,8,0.95)</code> with gold bottom border
Mobile	Hamburger menu, full-screen overlay with same nav links

UI-02: Button Styles

Button Type	CSS Specification	Usage
Primary (.btn-primary)	Gold gradient, black text, polygon clip-path, 16px 36px padding, hover: <code>translateY(-3px)</code> + gold shadow	Enrol Now, Join Session, Submit
Outline (.btn-outline)	Transparent bg, gold border (0.5 opacity), gold text, hover: gold bg at 7%	Secondary CTAs — Contact, Learn More
Mentor CTA (.mentor-btn)	Gold text, gold left-border (3px), padding 12px 20px, hover: gold bg at 8%	WhatsApp chat buttons in mentor section
Danger	Red (<code>#c0392b</code>) fill, white text	Delete, Suspend account (admin only)

UI-03: Form Inputs (Global Standard)

State	Specification
Default	Background: #131313, Border: 1px solid #232323, Text: #f0ede8, Montserrat 300, 14px
Focus	Border color: rgba(201,168,76,0.6), box-shadow: 0 0 0 3px rgba(201,168,76,0.1)
Error	Border color: #c0392b, error text below in #c0392b, Montserrat 300, 12px
Success	Border color: rgba(46,204,113,0.6)
Placeholder	Color: #888880 (--muted)
Label	Montserrat 700, 9px, letter-spacing: 3px, uppercase, color: --gold

UI-04: Student Dashboard Layout

- Sidebar navigation (collapsible on mobile) — dark black (#0e0e0e), gold accent icons
- Sidebar items: Dashboard, My Courses, Live Classes, My Profile, Support, Announcements
- Top bar: Logo, student name, notification bell, logout icon
- Main content area: white-card panels on #080808 background
- Gold rule dividers between sections
- Mobile: bottom tab navigation (5 icons)

UI-05: Admin Dashboard Layout

- Sidebar navigation — matches student sidebar style with additional admin items
- Admin sidebar items: Overview, Students, Content, Live Sessions, Announcements, Support, Trade Audits
- Admin badge in top bar showing 'MENTOR' or 'ADMIN' in gold
- Data tables: striped, sortable, with search bar and export button
- Stats cards: large Playfair Display numerals, gold color, with trend indicator arrows

UI-06: Super Admin Dashboard Layout

- Distinct visual differentiation from Admin — top bar has 'SUPER ADMIN' badge in red (#c0392b)
- Additional navigation sections: Revenue, Audit Log, Settings, Admin Management
- Charts and graphs using Chart.js or Recharts — styled with gold/dark theme
- Confirmation dialogs for all destructive actions (delete, suspend, role change)

UI-07: Mobile Responsiveness

Breakpoint	Behavior
320px – 639px (Mobile)	Single column layout, hamburger nav, bottom tab bar for dashboards, stacked cards

Breakpoint	Behavior
640px – 1023px (Tablet)	Two-column layouts, collapsible sidebar, optimized tables
1024px+ (Desktop)	Full sidebar visible, 3-column grids, wide tables with all columns

5. Non-Functional Requirements

5.1 Performance

NFR-ID	Requirement	Target
NFR-01	Page Load Time (First Contentful Paint)	< 2 seconds on 4G connection
NFR-02	API Response Time (95th percentile)	< 500ms for all REST API endpoints
NFR-03	Dashboard Load Time	< 3 seconds for authenticated dashboard
NFR-04	Payment Checkout Response	< 2 seconds to redirect to Cashfree
NFR-05	Concurrent Users	Handle 500 concurrent users without degradation
NFR-06	Database Query Performance	All indexed queries < 100ms

5.2 Security

NFR-ID	Requirement
NFR-07	All API endpoints require authentication (JWT) except public routes
NFR-08	Role-based access control (RBAC) enforced server-side — never trust client-side role claims
NFR-09	Passwords hashed with bcrypt (min 12 rounds)
NFR-10	JWT tokens stored in httpOnly, Secure, SameSite=Strict cookies
NFR-11	HTTPS enforced — HTTP redirects to HTTPS
NFR-12	Cashfree webhook signature verification on every webhook
NFR-13	Rate limiting: 100 req/min per IP on public routes, 5 failed logins = 15-min logout
NFR-14	Input sanitization and validation on all form fields (server-side)
NFR-15	CORS configured — only allow rajmudrafinitech.com origin
NFR-16	No sensitive data in frontend logs or client-visible responses
NFR-17	MongoDB — no raw queries; use Mongoose schema validation

NFR-ID	Requirement
NFR-18	Admin audit log stored separately and cannot be deleted by admin role

5.3 Usability & Accessibility

NFR-ID	Requirement
NFR-19	WCAG 2.1 AA minimum — contrast ratio 4.5:1 for normal text
NFR-20	All interactive elements keyboard-navigable (Tab order, focus visible)
NFR-21	Form error messages clear and specific (not just 'Invalid input')
NFR-22	Loading states shown for all async operations (skeleton loaders, spinners)
NFR-23	Success/failure toasts for all user actions (auto-dismiss in 4 seconds)
NFR-24	Bilingual support: English + Hindi — language switcher in student dashboard

5.4 Reliability & Maintainability

NFR-ID	Requirement
NFR-25	System uptime: 99.5% monthly SLA
NFR-26	Automated daily MongoDB backups — 30-day retention
NFR-27	Error monitoring via Sentry or equivalent
NFR-28	Graceful error handling — no raw stack traces visible to end users
NFR-29	Code split by route — lazy loading for dashboard modules
NFR-30	Environment variables for all secrets — no hardcoded credentials

6. Database Schema (MongoDB)

6.1 Collections Overview

Collection	Purpose
users	All users — students, admins, super admin
courses	Course structure and content (phases, materials, YouTube links)
payments	Cashfree payment records and invoice data
sessions	Scheduled Google Meet live sessions
tickets	Student support tickets and replies

Collection	Purpose
announcements	Admin announcements
tradeaudits	Trade submissions and admin feedback
auditlogs	Admin activity audit log
notifications	In-app notification queue

6.2 User Document Schema

Key fields for the users collection:

Field	Type	Description
_id	ObjectId	Auto-generated MongoDB ID
fullName	String	Student/admin full name
email	String	Unique, indexed, lowercase
whatsappNumber	String	10-digit Indian mobile number
password	String	bcrypt hashed — never returned in API responses
role	Enum	'student' 'admin' 'superadmin'
status	Enum	'pending_payment' 'active' 'suspended' 'pending_verification'
isEmailVerified	Boolean	Email confirmation status
interest	String	Selected interest at registration
language	Enum	'en' 'hi'
courseProgress	Object	{ phase1: boolean, phase2: boolean, ...phase5: boolean }
enrolledAt	Date	Payment completion timestamp
lastLogin	Date	Last successful login timestamp
createdAt / updatedAt	Date	Mongoose timestamps

6.3 Payment Document Schema

Field	Type	Description
_id	ObjectId	Auto-generated
orderId	String	RMF-{YYYY}-{8char} — unique
studentId	ObjectId	Ref: users

Field	Type	Description
amount	Number	39999 (INR paise stored as rupees)
currency	String	'INR'
status	Enum	'pending' 'success' 'failed' 'refunded'
cashfreeOrderId	String	Cashfree's order reference
paymentMethod	String	UPI / Card / Net Banking / EMI / Wallet
invoiceUrl	String	S3 or local URL of generated PDF invoice
paidAt	Date	Payment success timestamp
webhookPayload	Object	Raw Cashfree webhook data stored for audit

7. REST API Endpoints

7.1 Authentication Endpoints

Method	Endpoint	Auth Required	Description
POST	/api/auth/register	No	Student self-registration
POST	/api/auth/login	No	Student / Admin login — returns JWT cookies
POST	/api/auth/logout	Yes	Clear JWT cookies
POST	/api/auth/refresh	No (refresh cookie)	Get new access token
POST	/api/auth/forgot-password	No	Send password reset email
POST	/api/auth/reset-password/:token	No	Reset password with token
GET	/api/auth/verify-email/:token	No	Verify email address
POST	/api/auth/admin/login	No	Admin-specific login route

7.2 Student API Endpoints

Method	Endpoint	Description
GET	/api/student/dashboard	Dashboard overview data
GET	/api/student/profile	Own profile details
PUT	/api/student/profile	Update profile (name, language, avatar)
PUT	/api/student/change-password	Change password with current password verification

Method	Endpoint	Description
GET	/api/student/course	Course phases and unlocked content
GET	/api/student/sessions	Scheduled and past live sessions
GET	/api/student/payments	Own payment and invoice history
GET	/api/student/tickets	Own support tickets
POST	/api/student/tickets	Create new support ticket
POST	/api/student/tickets/:id/reply	Reply to own ticket
GET	/api/student/announcements	All announcements (paginated)
POST	/api/student/tradeaudit	Submit trade for audit
GET	/api/student/tradeaudit	Own trade audit submissions and feedback

7.3 Admin API Endpoints

Method	Endpoint	Description
GET	/api/admin/dashboard	Admin overview stats
GET	/api/admin/students	List all students with filters
GET	/api/admin/students/:id	Individual student details
PUT	/api/admin/students/:id/status	Activate / suspend student
GET	/api/admin/sessions	All sessions — past and upcoming
POST	/api/admin/sessions	Create new live session
PUT	/api/admin/sessions/:id	Edit session details
DELETE	/api/admin/sessions/:id	Cancel session (sends notifications)
GET	/api/admin/tickets	All support tickets with filters
PUT	/api/admin/tickets/:id	Update ticket status and add reply
POST	/api/admin/announcements	Create announcement
PUT	/api/admin/course/phase/:id	Update course phase content
GET	/api/admin/tradeaudits	List pending trade audits
PUT	/api/admin/tradeaudits/:id	Post feedback on trade audit

7.4 Super Admin API Endpoints

Method	Endpoint	Description
GET	/api/superadmin/dashboard	Full platform overview with revenue charts

Method	Endpoint	Description
GET	/api/superadmin/users	All users including admins
POST	/api/superadmin/users/admin	Create new admin account
PUT	/api/superadmin/users/:id/role	Change user role
DELETE	/api/superadmin/users/:id	Permanently delete user
GET	/api/superadmin/payments	All payment records
POST	/api/superadmin/payments/manual	Add manual payment entry
GET	/api/superadmin/auditlogs	Admin activity logs with filters
GET	/api/superadmin/settings	Platform settings
PUT	/api/superadmin/settings	Update platform settings
POST	/api/superadmin/maintenance	Toggle maintenance mode

7.5 Payment & Webhook Endpoints

Method	Endpoint	Description
POST	/api/payments/initiate	Create Cashfree order and return checkout token
GET	/api/payments/status/:orderId	Check payment status (polling fallback)
POST	/api/webhooks/cashfree	Cashfree webhook — signature verified, updates DB

8. External Integrations

8.1 Cashfree Payment Gateway

Parameter	Value
Integration Mode	Cashfree JS SDK (web checkout) — seamless payment experience
API Version	2023-08-01 (latest stable)
Webhook Event	PAYMENT_SUCCESS, PAYMENT_FAILED — verified with X-Webhook-Signature header
Test Credentials	Sandbox environment for development, production keys in env vars
Documentation	https://docs.cashfree.com/
Supported UPI Apps	Google Pay, PhonePe, Paytm, BHIM, and all UPI apps

8.2 Google Meet

Integration Type	Description
Method	Link delivery — no API integration required initially
Admin Action	Admin creates Google Meet session externally, pastes link into Admin Dashboard
Link Storage	Stored in sessions collection in MongoDB
Link Reveal	Link shown to students only when session is within 15 minutes of start time
Link Expiry	Session marked as completed after scheduled end time; link no longer accessible
Future Enhancement	Google Calendar API + Google Meet API integration in Phase 2

8.3 YouTube

Integration Type	Description
Method	YouTube Unlisted Video Embed — iframe embedded on course content pages
Admin Action	Admin uploads video to YouTube as Unlisted, copies embed URL into Admin Dashboard
Security	URL stored server-side, embedded only in authenticated pages — not exposed in public HTML
Player Controls	Standard YouTube controls — no autoplay
Mobile	Responsive 16:9 iframe container

8.4 WhatsApp Business

Usage	Message Template
Navigation WhatsApp icon	"Hi Rajmudra Fintech, I'm interested in your trading programme. Can you share more details?"
Hero CTA	"Hi Rajmudra Fintech, I want to know more about your courses and membership plans."
Mentor — Vishal	"Hi, I'm interested in Rajmudra Fintech. Can you share more details about the courses?" → wa.me/917020408147
Mentor — Nilesh	"Hi, I'm interested in Rajmudra Fintech. Can you share more details about the courses?" → wa.me/919370473645
Contact — keyword	"TRADER" → pre-filled in WhatsApp to trigger automated syllabus delivery

8.5 Email (Transactional)

Event	Template
Email Verification	Verify email link — expires 24 hours
Welcome Email	Post-payment enrolment welcome with dashboard link
Payment Success	Payment confirmation with invoice PDF attached
Password Reset	Reset link — expires 1 hour
Live Session Reminder	Sent 1 hour before each session with Meet link reminder
Session Cancellation	Notifies students when admin cancels a session
Ticket Update	Notifies student when admin replies or resolves a ticket
Announcement	Important announcements blast to all active students

9. Deployment Architecture

9.1 Recommended Architecture

Component	Recommended Service	Notes
Frontend (React)	Vercel / Netlify	CI/CD on push to main branch
Backend (Node/Express)	Railway / Render / DigitalOcean	Node.js server, env vars configured
Database (MongoDB)	MongoDB Atlas (M10+ cluster)	Automated backups, VPC peering
File Storage	AWS S3 or Cloudinary	Course PDFs, trade audit images, invoices
Email	SendGrid / Resend	Transactional email delivery
SSL/TLS	Let's Encrypt (auto)	Enforced on all endpoints
CDN	Cloudflare	DDoS protection, caching, DNS
Domain	rajmudrafintech.com	Primary domain — dashboard at app.rajmudrafintech.com
Monitoring	Sentry + UptimeRobot	Error tracking + uptime alerts

9.2 Environment Variables

All sensitive configuration must be stored as environment variables. Key variables include:

- MONGODB_URI — MongoDB Atlas connection string
- JWT_SECRET — Access token signing secret (min 64 chars random)
- JWT_REFRESH_SECRET — Refresh token signing secret
- CASHFREE_APP_ID — Cashfree API key
- CASHFREE_SECRET_KEY — Cashfree secret key
- CASHFREE_WEBHOOK_SECRET — For signature verification
- SENDGRID_API_KEY — Email service API key
- AWS_S3_BUCKET / AWS_ACCESS_KEY / AWS_SECRET — File storage
- NODE_ENV — 'development' | 'production'
- FRONTEND_URL — For CORS configuration

10. Development Phases & Timeline

Phase	Features	Estimated Duration
Phase 1 — Core	MERN setup, Auth (register/login/JWT), Public website React rebuild, Cashfree payment, Student dashboard basic, Admin dashboard basic	6-8 weeks
Phase 2 — Content & Classes	Course content management, YouTube embed delivery, Live session scheduling (Google Meet link), Trade audit submission, Announcements	4-6 weeks
Phase 3 — Super Admin	Super admin dashboard, Full audit log, Revenue analytics, Admin management, Platform settings	3-4 weeks
Phase 4 — Polish & Launch	Email templates, Hindi language support, Mobile optimization, Performance optimization, Security audit, UAT testing, Go-live	3-4 weeks
Phase 5 — Enhancement	Cashfree refund API, Google Calendar integration, Real-time ticker API, Advanced analytics, 2FA for admins	Ongoing

11. Acceptance Criteria

AC-ID	Criteria	Priority
AC-01	Static website fully rebuilt in React with identical design, typography, and all existing content	Critical
AC-02	Student registration, email verification, and login flow works end-to-end	Critical
AC-03	Cashfree payment of INR 39,999 processes successfully and unlocks student dashboard	Critical
AC-04	Student dashboard shows course phases, live session schedule, and support tickets	Critical

AC-ID	Criteria	Priority
AC-05	YouTube lectures embedded correctly — only visible to authenticated active students	Critical
AC-06	Google Meet link visible to students only within 15 minutes of session start	High
AC-07	Admin can schedule sessions, manage students, post announcements, reply to tickets	Critical
AC-08	Super Admin can view all revenue, manage all users, view admin audit log	Critical
AC-09	All forms have server-side validation with clear, specific error messages	High
AC-10	JWT authentication enforced on all protected routes — no unauthorized access possible	Critical
AC-11	Cashfree webhook correctly updates student status on payment success/failure	Critical
AC-12	Website is fully responsive on mobile (320px+), tablet, and desktop	High
AC-13	HTTPS enforced, CORS restricted to rajmudrafinitech.com	Critical
AC-14	Password reset and email verification emails delivered within 60 seconds	High
AC-15	All WhatsApp CTA links open with correct pre-filled messages	Medium
AC-16	PDF invoice auto-generated on payment and emailed to student	High
AC-17	Admin activity audit log records all admin actions with timestamp and IP	High
AC-18	Page load time under 3 seconds on 4G connection (measured with Lighthouse)	High

12. Glossary

Term	Definition
MERN	MongoDB, Express.js, React.js, Node.js — full-stack JavaScript framework
JWT	JSON Web Token — stateless authentication mechanism
Cashfree	Indian payment gateway supporting UPI, cards, net banking, EMI, wallets
Google Meet	Video conferencing tool used for live trading sessions
YouTube Unlisted	YouTube video accessible only via direct link — not searchable publicly
Webhook	HTTP callback triggered by Cashfree to notify payment status changes
RBAC	Role-Based Access Control — permission model based on user roles

Term	Definition
FEMA	Foreign Exchange Management Act — Indian law governing forex transactions
IST	Indian Standard Time — UTC+5:30
VTP	Vishal Tompe Patil — Founder & Mentor, Rajmudra Fintech
NTP	Nilesh Thore Patil — Founder & Mentor, Rajmudra Fintech
Phase	A structured module of the Professional Forex Mastery course (5 total)
Trade Audit	Review service where mentors provide feedback on student's actual trades
Super Admin	Platform owner with unrestricted system access

— *End of Document* —

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